AFTERWARD

Clearer discharge documents. Fewer unresolved questions.

Understand what is there. Find what is missing.
Keep the remaining conversation connected.

Google Gemma on AMD | Health and Wellbeing

Live demo: afterward-demo.up.railway.app

THE PROBLEM

A correct letter can still leave a practical gap.

The patient needs an answer. The ward needs the context.

"Follow the attached rehabilitation programme."

But the attachment is not in the packet.
Or the instruction is present, but unfamiliar.

8.05 million

discharges recorded in Italy in 2024 Episodes across care regimes and outcomes.

14.5 minutes

median summary completion in one hospital study of 7,185 records.

Sources: ISS / SDO 2024; Kleinig et al., 2025. Context only: neither figure measures AfterWard impact.

THE WORKFLOW

Start where the documents already are.

One case connects source documents, AI suggestions and the remaining questions.

01 Before discharge

The ward checks the packet, adds missing documents and shares the reviewed bundle.

02 At home

The patient reads simpler wording and looks for answers in their documents.

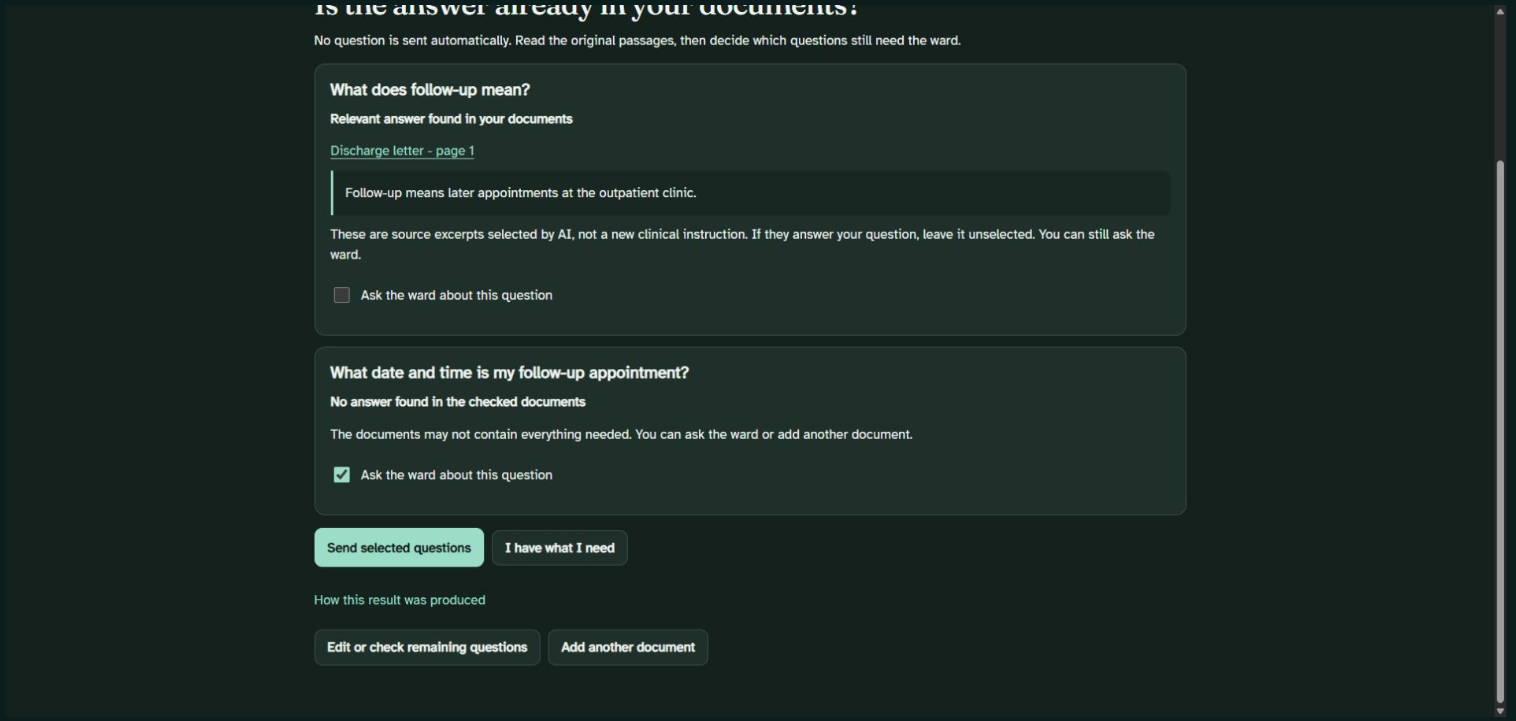
03 Ask the ward

Selected questions go to staff. Follow-ups stay together. The patient marks resolution.

The key distinction: a document held by the ward is not yet a document received by the patient.

Ask the documents before asking the ward.

The patient chooses what still needs a human reply.



Already present

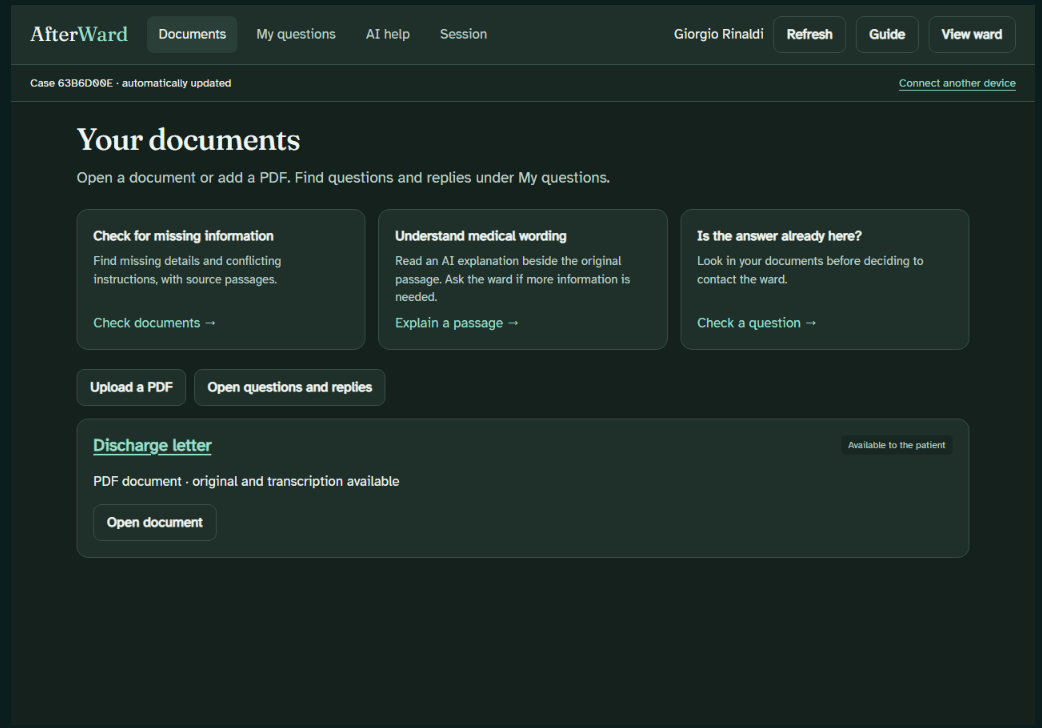
Read the cited passage. Send only if you still need clarification.

Still missing

Forward only the selected question, in context.

AI help is part of the document workflow.

Check the packet, explain a passage, or look for an existing answer.



Staff can reply directly. Supporting documents and AI drafts are optional.

TECHNOLOGY

Gemma handles the images and the text.

Actual API calls through AMD. Source checks and explicit human actions surround the model.

Task	Reasoning	Publication / control
Scanned PDF transcription	None	User checks every page
Missing or conflicting instructions	Medium	Facts -> findings -> conditional conflict check
Answer in patient documents	Medium	Exact excerpts; patient selects what to send
Ward reply draft	Medium	Staff edits and sends
Prepared clinical explanations	Medium	Shared cache; direct patient access

Python + SQLite + PDFium + Railway | Document revisions, role filtering, bounded requests and explicit failures

MEASURED EVIDENCE

Strong ward results; patient false alerts remain.

Frozen English synthetic documents. Google Gemma medium. Errors retained.

Task / pipeline	Cases	Exact cases	Precision	Recall
Patient document gaps / v7	100	73.00%	64.15%	94.44%
Ward request handling / v2	100	97.00%	98.31%	96.67%

Exact = the whole expected finding set matches.
Precision = correct findings among alerts. Recall = expected findings recovered.

Includes 4 patient acquisition controls and 20 deterministic ward gates. Related variants are clustered.

Not clinical validation or an evaluation of every integrated UI feature.

BUSINESS VALUE

Count the staff time released by each case.

Illustrative annual ward scenario. The assumptions must be tested in a pilot.

Assumption / calculation	Illustrative amount
10,000 cases x 2 net min x EUR30/hour	EUR10,000 capacity value
AI EUR0.02 + other EUR0.10 per case	EUR1,200 variable budget
Fixed operating / integration budget	EUR5,000 per year
Capacity value minus total budget	EUR3,800 per year

1.24 min

net time saved per case to reach this scenario's break-even point.

Buyer hypothesis: hospital discharge teams. Pilot metric: net minutes per case, including review and false alerts.

Two contributions, one economical workflow.

Gemma supplies the model capability. AMD supplies the live inference service.

Model and workflow

Bounded source text, task-specific prompts, selective calls and validated outputs keep consumption visible and controllable.

Provider execution

The demo runs real image and text requests through AMD. Shared capacity and free quota are monitored; failures stay visible.

10k input + 2k output tokens: about $0.00168 at the CoreWeave / OpenRouter reference tariff.

Observed 17 Oct 2026. OCR and retries add consumption. AMD quota coefficients do not establish a paid-price saving.

NEXT STEP

A controlled ward pilot with clear success tests.

The prototype demonstrates the workflow. A real service requires a separate deployment boundary.

Value	Net staff minutes per discharge and clarification request; patient understanding.
Quality	Independent clinical review; false alerts, missed issues and source fidelity.
Operations	Verified staff/patient/delegate identity, appropriate data terms and retention controls.

The present hosted demo uses invented examples and a development inference endpoint.

The handoff is the product.

Document possession, document delivery and patient understanding are different states.

What AfterWard adds

A ward-first check before discharge.
Answers limited to what the patient received.
Selective clarification, threaded follow-up
and patient-owned resolution.

Foundations disclosed

Selected FHIR, schema and audit utilities come from the author's MIT-licensed TrustedRisk project.

The runnable demo accepts PDFs; a live EHR connector is future work.

TRY IT

Follow one question from source to resolution.

Open the guided cases and inspect the public code.

afterward-demo.up.railway.app

1 Check a packet before sharing it.
2 Find an existing answer before contacting staff.
3 Read a scan, then keep a clarification connected.

github.com/AlessandroFlati/AfterWard